

UFHE – Product Requirements Document (PRD)
& Complete Data Source Directory
PART A: PRODUCT REQUIREMENTS DOCUMENT
1. DOCUMENT HEADER
text

PRODUCT REQUIREMENTS DOCUMENT
Product Name : Urban Flooding & Hydrology Engine (UFHE) Version : 1.0 Document Owner: [Your Name / Team Lead] Last Updated : [Current Date] Status : DRAFT → REVIEW → APPROVED Stakeholders: — Municipal Corporation (Primary Customer) — Ward Engineers (Primary Users) — City Commissioner / Mayor (Executive Sponsor) — Disaster Management Authority (Partner) — Meteorological Department (Data Provider) — Citizens (End Beneficiaries)

2. PRODUCT VISION & OBJECTIVES
2.1 Vision Statement
text

"Transform urban flood management from REACTIVE crisis response to PROACTIVE, data-driven prevention by giving every city ward a living, breathing flood intelligence system that predicts waterlogging before the first drop falls."

2.2 Problem Statement
text

THE PROBLEM (Quantified)
CURRENT STATE: • Indian cities lose ₹15,000-50,000 Cr annually to urban flooding (NDMA estimates) • Average response time after waterlogging: 4-12 hours • 70% of drainage infrastructure condition: UNKNOWN • Flood prediction: based on "gut feel" + last year's memory • Resource deployment: ad-hoc, reactive, politically driven • Zero quantified ward-level readiness benchmarking ROOT CAUSES: • No unified spatial view of terrain + drainage + rainfall • Drainage capacity data exists in paper registers, if at all • No systematic method to identify ALL vulnerable micro-zones • Pre-monsoon preparation is checklist-based, not risk-based • No feedback loop: same areas flood year after year DESIRED STATE: • Every 50m × 50m patch of the city has a flood risk score • Ward engineers know EXACTLY which spots will flood and WHY • Resources pre-positioned HOURS before rain, not after • Commissioner has a single dashboard: "Are we ready? Where are we NOT ready?" • Year-over-year measurable improvement in flood outcomes

2.3 Success Metrics (OKRs)
text

OBJECTIVE 1: Accurate Flood Prediction

- KR1: Identify 2,500+ micro-hotspots with >80% recall
- KR2: Predict flood depth within $\pm 20\text{cm}$ RMSE
- KR3: Provide 3+ hours lead time for resource deployment
- KR4: False negative rate < 15% (missed real floods)

OBJECTIVE 2: Proactive Readiness

- KR1: 100% wards scored on PMRS before every monsoon
- KR2: Average PMRS improves by 15 points year-over-year
- KR3: 90% of critical hotspots have pre-positioned resources
- KR4: Response time reduced from 6hrs to <2hrs

OBJECTIVE 3: Platform Adoption

- KR1: 100% ward engineers trained and using weekly
- KR2: Commissioner reviews dashboard at least 2x/week during monsoon
- KR3: Field teams using mobile app for 80% of deployments
- KR4: System uptime > 99.5% during monsoon months

OBJECTIVE 4: Measurable Impact

- KR1: 30% reduction in waterlogging complaints (Year 1)
- KR2: 50% reduction in flood-related road closures
- KR3: Zero flood-related deaths in mapped hotspots
- KR4: ₹X Cr estimated damage prevented (economic model)

3. USER PERSONAS & JOURNEYS

3.1 Persona Definitions

text

PERSONA 1: "Commissioner Sharma"

Role: Municipal Commissioner / Additional Commissioner

Age: 50-58 | IAS Officer

Tech Comfort: Medium (uses email, WhatsApp, basic dashboards)

GOALS:

- Single-screen city-wide flood readiness overview
- Know which wards need intervention BEFORE monsoon
- Defend budget allocation with data
- Avoid media embarrassment from preventable flooding
- Compare ward performance objectively

PAIN POINTS:

- Gets 50+ WhatsApp messages during heavy rain
- No single source of truth
- Can't tell which complaints are genuine emergencies
- Ward engineers blame rain intensity; no way to verify

KEY INSIGHT: Needs SIMPLICITY. Traffic-light colors.

Wants to open one screen and know: "Where are we screwed?"

PERSONA 2: "Engineer Priya"

Role: Ward Engineer / Assistant Engineer

Age: 30-45 | Civil Engineering background

Tech Comfort: Medium-High (uses GIS apps, smart phone native)

GOALS:

- Know exactly WHICH drains to desilt first (priority order)
- Justify equipment requests with flood risk data

- Get early warning to pre-position pumps
- Track her ward's readiness progress
- Reduce complaint volume through prevention

PAIN POINTS:

- Manages 50+ km of drainage, can't inspect everything
- Limited pumps (3-4) for 15+ potential flood spots
- Complaint data is unstructured and overwhelming
- No historical analysis of which interventions worked

KEY INSIGHT: Needs ACTIONABLE details. "Do THIS at THAT location by THIS date." Prioritized checklist.

PERSONA 3: "Supervisor Raju"

Role: Field Supervisor / Pump Operator

Age: 35-50 | ITI/Diploma background

Tech Comfort: Low-Medium (smartphone, WhatsApp, basic apps)

GOALS:

- Know where to go and what to carry BEFORE rain starts
- Navigate to exact flood location quickly
- Report status back without writing lengthy reports
- Know when it's safe to leave / move to next location

PAIN POINTS:

- Gets contradictory instructions from multiple sources
- Arrives at location without right equipment
- No systematic way to report ground conditions
- Works 16+ hour shifts during heavy rain, low visibility

KEY INSIGHT: Needs SIMPLE MOBILE APP. Maps, directions, one-tap status updates. Works offline in poor connectivity.

PERSONA 4: "Citizen Ananya"

Role: Resident of flood-prone area

Age: 25-60 | Various backgrounds

Tech Comfort: Varies

GOALS:

- Know if her area will flood today
- Report waterlogging and get response
- See that government is taking action
- Plan her commute around flood-prone routes

KEY INSIGHT: Needs TRANSPARENCY. Public flood risk map. SMS/notification alerts. Complaint tracking.

3.2 User Journey Maps

text

JOURNEY: "Commissioner Sharma – Pre-Monsoon Review" (May)

Step 1: LOGIN → Dashboard

↓

Step 2: SEE city-wide PMRS summary

"Average: 58/100 | Grade C | 12 wards below 40"

↓

Step 3: CLICK worst-performing wards → see specific gaps

"Ward 47: Desilting 20% complete. 3 pumps non-functional."

- ↓
- Step 4: GENERATE "Pre-Monsoon Readiness Report" (PDF)
Auto-generated with ward rankings, gap analysis,
recommended budget allocation
- ↓
- Step 5: SHARE report with ward engineers with deadlines
"Ward 47: Complete desilting by June 5. Fix pumps by June 1."
- ↓
- Step 6: TRACK progress weekly on dashboard
Watch PMRS scores improve as tasks are completed
- ↓
- Step 7: BY JUNE 15: All wards at Grade B or above = READY

JOURNEY: "Engineer Priya – Active Rainfall Event" (July)

- Step 1: RECEIVE push notification at 10:00 AM
"⚠️ ORANGE ALERT: 65mm/hr rainfall predicted for your ward in next 3 hours. 8 hotspots at HIGH risk."
- ↓
- Step 2: OPEN app → see ward map with 8 highlighted hotspots
Each hotspot shows: predicted depth, timing, affected pop.
- ↓
- Step 3: VIEW auto-generated deployment plan
"Pump A → Hotspot #3 (80cm predicted) by 11:00 AM
Pump B → Hotspot #7 (55cm predicted) by 11:00 AM
Barrier → Hotspot #1 (underpass, close by 10:30 AM)"
- ↓
- Step 4: APPROVE / MODIFY plan → sends instructions to field teams
- ↓
- Step 5: MONITOR real-time during rain
See actual sensor readings vs predictions
Receive field team status updates
- ↓
- Step 6: POST-EVENT: Review performance
"6 of 8 hotspots correctly predicted.
Hotspot #3: predicted 80cm, actual 65cm. ✓
Hotspot #5: predicted 20cm, actual 45cm. ✗ Investigate."

4. FUNCTIONAL REQUIREMENTS

4.1 Feature Specification – Module Breakdown text

MODULE 1: GIS MAP ENGINE (Core Visualization) Priority: P0 (Must Have)
F1.1 Interactive web map with pan, zoom, search - Base layer: satellite imagery + street map toggle - Coordinate display, measurement tools - Print/export map as image or PDF F1.2 Layer management panel - Toggle ON/OFF: ward boundaries, micro-basins, drainage network, flood hotspots, DEM hillshade, land use, building footprints, water bodies - Opacity slider per layer - Legend auto-updates with visible layers F1.3 Thematic mapping - Choropleth: wards colored by PMRS score - Graduated symbols: hotspots sized by severity - Heatmap: flood risk density

- Drainage network colored by capacity utilization %

F1.4 Spatial query tools

- Click any location → show flood risk, elevation, nearest drain, ward info, historical floods
- Draw polygon → get aggregate statistics
- Buffer query: "Show all hotspots within 500m of this hospital"

F1.5 3D terrain visualization

- DEM-based 3D view (Deck.gl TerrainLayer)
- Flood depth overlay on 3D terrain
- Drainage pipe profile (cross-section view)

ACCEPTANCE CRITERIA:

- ✓ Map loads within 3 seconds on 4G connection
- ✓ Smooth interaction with 2,500+ hotspot markers
- ✓ Works on Chrome, Firefox, Edge (latest 2 versions)
- ✓ Responsive: usable on tablet (1024px+)

MODULE 2: FLOOD HOTSPOT IDENTIFICATION ENGINE

Priority: P0 (Must Have)

F2.1 Micro-basin delineation

- Auto-generate 2,500+ micro-basins from DEM
- Each basin: unique ID, area, pour point, avg slope, avg elevation, elevation range
- Visualize basin hierarchy (which drains into which)

F2.2 Flood Vulnerability Index (FVI) computation

- Compute FVI for each micro-basin
- Components: historical frequency, elevation depression, drainage deficit, imperviousness, TWI, upstream area
- Configurable weights (admin can adjust)
- FVI re-computation on data updates

F2.3 Hotspot classification and ranking

- Classify: Very High / High / Moderate / Low / Very Low
- Global rank: 1 to N across city
- Ward-level rank: within each ward
- Filter / search / sort by any attribute

F2.4 Scenario-based flood simulation

- Input: rainfall amount (mm) + duration (hours)
- Output: which basins flood, predicted depth, volume
- Pre-built scenarios: 2yr, 5yr, 10yr, 25yr, 50yr return period storms
- Custom scenario: user enters any rainfall value

F2.5 Drainage deficit analysis

- Per micro-basin: required capacity vs available capacity
- Deficit map: red zones = demand >> supply
- "What-if": simulate adding new drain pipe → see impact on deficit

ACCEPTANCE CRITERIA:

- ✓ 2,500+ unique micro-basins generated
- ✓ FVI computed for 100% of basins
- ✓ Scenario simulation runs in < 30 seconds
- ✓ Historical validation: >80% recall on past floods

MODULE 3: PRE-MONSOON READINESS SCORE (PMRS)
Priority: P0 (Must Have)

F3.1 PMRS computation engine

- Compute 4-component score per ward
- Infrastructure (30%) + Vulnerability (25%) + Response Prep (25%) + Historical Performance (20%)
- Auto-pull data where available; manual entry forms for field-collected data

F3.2 PMRS data collection forms

- Web forms for ward engineers to input:
 - Desilting progress (km completed / total km)
 - Pump status (operational / under repair / defunct)
 - Equipment inventory
 - Team strength and training status
- Photo upload for verification
- Approval workflow (engineer → zonal officer)

F3.3 PMRS dashboard

- City-wide summary: avg score, grade distribution
- Ward ranking table (sortable by total & component)
- Spider/radar chart per ward (4 components)
- Trend chart: PMRS over time (month-on-month)
- Comparison view: select 2-5 wards side-by-side

F3.4 Gap analysis & action items

- Auto-generate ward-specific action items from gaps
- Priority ordering by impact on PMRS improvement
- Assignment to responsible officer
- Due dates and progress tracking
- Notification/escalation on overdue items

F3.5 Pre-Monsoon Readiness Report (auto-generated)

- PDF report per ward and city-wide
- Executive summary, scores, gaps, recommendations
- Comparison with previous year
- Budget estimation for gap closure

ACCEPTANCE CRITERIA:

- ✓ PMRS computed for 100% of wards
- ✓ Score update reflects within 24 hours of data entry
- ✓ Report generation < 60 seconds
- ✓ Action items auto-generated with 90%+ relevance

MODULE 4: RAINFALL ANALYSIS & FORECASTING
Priority: P0 (Must Have)

F4.1 Historical rainfall analysis

- IDF curves for the city (auto-computed from data)
- Return period analysis (2, 5, 10, 25, 50, 100 yr)
- Monthly/seasonal rainfall statistics
- Trend analysis (is rainfall increasing over decades?)
- Extreme event catalog

F4.2 Real-time rainfall ingestion

- Pull from AWS stations every 15 minutes
- Pull from IMD radar/satellite every 30 minutes

- Spatial interpolation to micro-basin level
- Running accumulation display (1hr, 3hr, 6hr, 24hr)
- Comparison with IDF: "Current event = X-yr storm"

F4.3 Forecast integration

- IMD district/city forecast (text + QPF)
- GFS/ECMWF model output (if accessible)
- Nowcasting: extrapolate radar for 0-3 hours
- Confidence level displayed with each forecast

F4.4 Alert generation

- Threshold-based alerts when rainfall exceeds design capacity of drainage
- Alert levels: GREEN → YELLOW → ORANGE → RED
- Delivery: in-app, SMS, email, WhatsApp API
- Customizable thresholds per ward (different areas have different capacities)

ACCEPTANCE CRITERIA:

- ✓ Rainfall data refreshes every 15 minutes during events
- ✓ Alerts delivered within 5 minutes of threshold breach
- ✓ IDF curves validated against published literature
- ✓ Forecast lead time ≥ 3 hours

MODULE 5: PREDICTIVE ENGINE (ML-Enhanced)

Priority: P1 (Should Have)

F5.1 Physics-ML hybrid flood prediction

- Real-time: given current + forecast rainfall, predict which hotspots will activate and depths
- Update predictions every 15-30 minutes during event
- Confidence intervals on all predictions
- Explainability: top 3 factors driving each prediction

F5.2 Model management

- Model versioning and A/B testing
- Performance metrics dashboard (accuracy over time)
- Automated retraining pipeline (post-monsoon)
- Manual trigger for retraining with new data

F5.3 Temporal prediction

- Timeline view: "Hotspot X activates at T+45min, peaks at T+90min, clears at T+180min"
- Animated flood progression map

ACCEPTANCE CRITERIA:

- ✓ Prediction runs within 60 seconds for entire city
- ✓ RMSE < 20cm for flood depth predictions
- ✓ Recall > 85% for hotspot activation

MODULE 6: RESOURCE DEPLOYMENT MANAGER

Priority: P1 (Should Have)

F6.1 Resource inventory management

- Track: portable pumps, dewatering machines, barriers, boats, generators, sandbags, response teams
- Location tracking (which depot / which ward)
- Status: available / deployed / under maintenance

- F6.2 Optimized deployment suggestions
- Given forecast + hotspot predictions, suggest optimal resource allocation
 - Minimize total expected damage / response time
 - Respect constraints: travel time, capacity, shifts
- F6.3 Deployment tracking
- Real-time tracking of deployed resources (GPS)
 - Status updates from field (arrived, operational, done)
 - Time-stamped audit trail
- ACCEPTANCE CRITERIA:
- ✓ Deployment plan generated within 5 minutes of forecast
 - ✓ 90%+ resource utilization during events

MODULE 7: MOBILE APP (Field Operations)
Priority: P1 (Should Have)

- F7.1 Field team features
- Assigned hotspots with navigation
 - One-tap status updates (en route / arrived / pumping / cleared)
 - Photo/video capture with auto-geotagging
 - Water depth measurement input
 - Offline mode with data sync when connected
- F7.2 Citizen reporting (public-facing)
- Report waterlogging with photo + location
 - Auto-categorize severity from photo (future: CV)
 - Track complaint status
 - View nearby active hotspots / road closures
- F7.3 Alerts & notifications
- Push notifications for assigned deployments
 - Area-based flood alerts for citizens
 - Offline alert caching
- ACCEPTANCE CRITERIA:
- ✓ App works offline for core functions
 - ✓ Photo upload < 10 seconds on 3G
 - ✓ Battery usage < 5% per hour of active use

MODULE 8: REPORTING & ANALYTICS
Priority: P1 (Should Have)

- F8.1 Standard reports
- Pre-Monsoon Readiness Report (annual)
 - Post-Event Analysis Report (per event)
 - Monthly Monsoon Summary
 - Annual Performance Review
 - Ward Comparison Report
- F8.2 Analytics dashboards
- Trend analysis: floods over years
 - Correlation: rainfall vs flood events
 - Infrastructure ROI: investment vs flood reduction
 - Response time analytics

- Prediction accuracy over time

F8.3 Data export

- Export any dataset as CSV, Shapefile, GeoJSON, KML
- API access for integration with other city systems
- Scheduled report delivery (email)

ACCEPTANCE CRITERIA:

- ✓ Reports generated in < 60 seconds
- ✓ All data exportable in at least 2 formats
- ✓ API response time < 2 seconds for standard queries

MODULE 9: ADMINISTRATION & CONFIGURATION

Priority: P0 (Must Have)

F9.1 User management

- Roles: Super Admin, City Admin, Zonal Officer, Ward Engineer, Field Staff, Viewer, Public
- Permission matrix per module and per ward
- SSO integration (if city has identity system)
- Audit logging of all user actions

F9.2 Configuration management

- Alert thresholds (customizable per ward)
- PMRS weights (adjustable)
- FVI weights (adjustable)
- Notification channels and recipients
- Map styling and layer defaults

F9.3 Data management

- Upload interface for new GIS data layers
- Data quality dashboard
- Backup and restore
- Data versioning (track changes to spatial data)

ACCEPTANCE CRITERIA:

- ✓ Role-based access enforced on all endpoints
- ✓ Configuration changes reflected without app restart
- ✓ Full audit trail searchable for 2+ years

4.2 Feature Priority Matrix text

MoSCoW PRIORITIZATION		
PRIORITY	FEATURES	PHASE
MUST HAVE (P0)	GIS Map with layer management Micro-basin delineation FVI computation & hotspot map PMRS scoring & dashboard Basic drainage capacity analysis Historical rainfall analysis User management & auth Static scenario simulation	Phase 1 (MVP) Months 1-4
SHOULD HAVE	Real-time rainfall ingestion ML prediction engine Alert system (SMS/email/push) Resource deployment optimizer	Phase 2 Months 4-8

(P1)	Mobile app (field teams) Post-event analysis reports Temporal flood progression	
COULD HAVE (P2)	3D terrain visualization Citizen reporting module Social media NLP integration What-if infrastructure simulator Economic impact estimator Climate change projections	Phase 3 Months 8-12
WON'T HAVE (now) (P3)	IoT sensor integration Digital twin drainage simulation Automated drone survey pipeline Multi-city federation AR field visualization	Future Releases

5. NON-FUNCTIONAL REQUIREMENTS

text

NON-FUNCTIONAL REQUIREMENTS
PERFORMANCE - Map initial load: < 3 seconds (fiber) / < 6 seconds (4G) - Map interaction (pan/zoom): < 200ms response - Hotspot layer render (2,500+ points): < 1 second - PMRS computation (all wards): < 30 seconds - Scenario simulation: < 30 seconds - API response (standard queries): < 2 seconds - Report generation: < 60 seconds - ML prediction (full city): < 60 seconds SCALABILITY - Support city area up to 1,500 km² - Up to 10,000 micro-basins - Up to 500 concurrent dashboard users - Up to 10,000 mobile app users 10 years of historical data storage - Real-time data ingestion from 200+ sensors AVAILABILITY - Overall uptime: 99.5% - During monsoon (Jun-Sep): 99.9% (critical period) - Planned maintenance: only during dry season - RTO (Recovery Time): < 1 hour - RPO (Recovery Point): < 15 minutes SECURITY - HTTPS everywhere (TLS 1.3) - JWT-based authentication - Role-based access control (RBAC) - API rate limiting - SQL injection prevention (parameterized queries) - Data encryption at rest (AES-256) - Audit logging of all data modifications - VAPT (Vulnerability Assessment) annually - Compliance with IT Act 2000 / data localization USABILITY - Mobile-responsive (tablet minimum for web; phone for app) - Multi-language: English + local language (Hindi/regional) - Accessibility: WCAG 2.1 AA compliance

- Color-blind safe color palettes for maps
- Maximum 3 clicks to reach any key information
- Contextual help tooltips on all technical terms
- Onboarding tutorial for first-time users

INTEGRATION

- RESTful API with OpenAPI 3.0 specification
- GeoJSON / WMS / WFS standard GIS interfaces
- Webhook support for external alert systems
- SMS gateway integration (Twilio / govt SMS gateway)
- WhatsApp Business API (optional)
- Email (SMTP)

DATA RETENTION

- Real-time sensor data: 2 years at full resolution, then aggregated to hourly for archival
- Predictions & alerts: indefinite (for accuracy tracking)
- User actions & audit logs: 5 years
- GIS layers: versioned, all versions retained
- Reports: indefinite

DEPLOYMENT

- Containerized (Docker)
- Infrastructure as Code (Terraform)
- CI/CD pipeline with automated testing
- Blue-green deployment (zero downtime)
- Environment: Dev → Staging → Production
- Cloud-agnostic (deployable on AWS / Azure / GovCloud)

6. INFORMATION ARCHITECTURE & WIREFRAMES

6.1 Navigation Structure

text

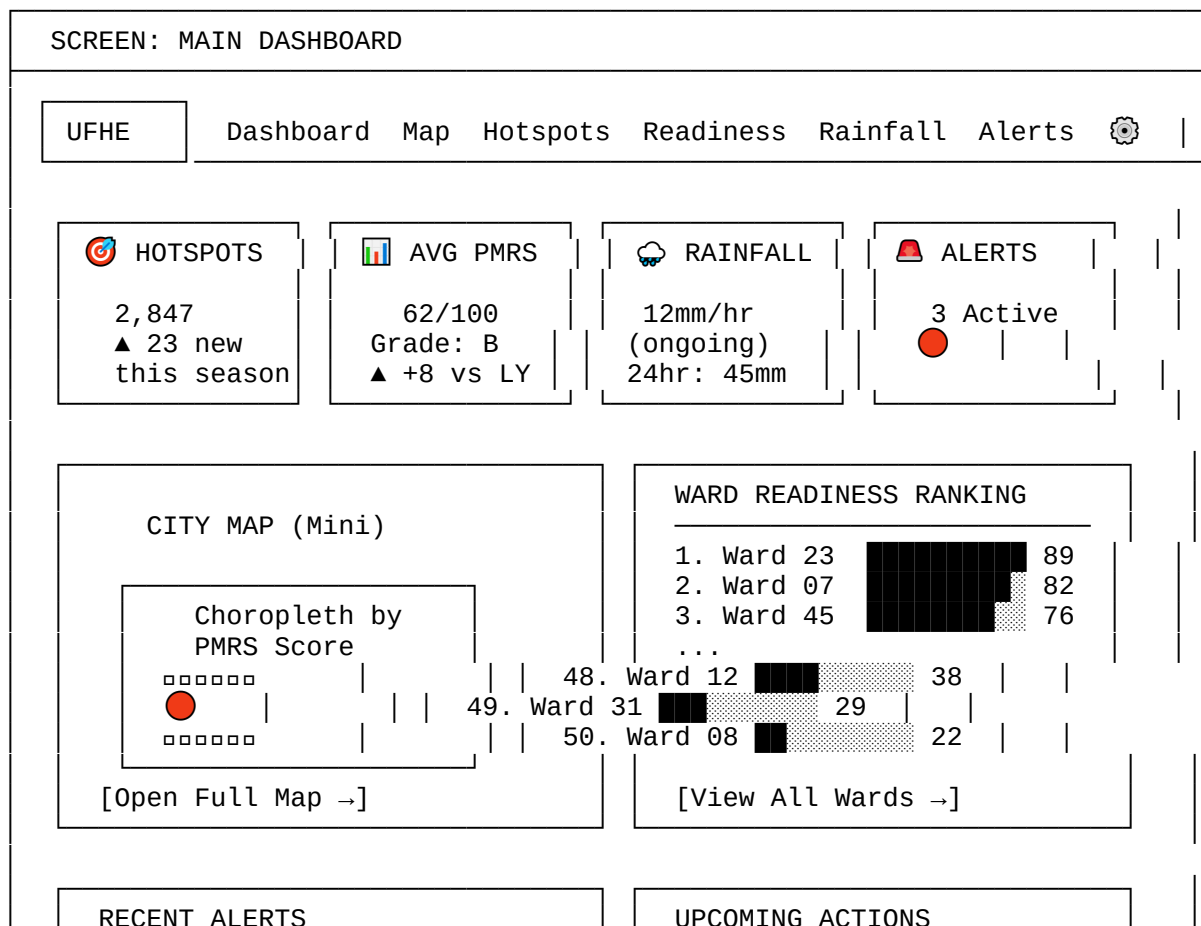
UFHE Web Application

-  Dashboard (Landing Page)
 - City Overview Panel
 - Key Metrics Summary
 - Active Alerts Feed
 - Quick Actions
-  Map Explorer
 - Layer Panel (left sidebar)
 - Map Canvas (center)
 - Feature Info Panel (right sidebar, on click)
 - Tools Bar (top: measure, draw, search, print)
 - Legend (bottom-left)
-  Flood Hotspots
 - Hotspot List View (table with filters)
 - Hotspot Detail View (per hotspot deep-dive)
 - Hotspot Map View (focused hotspot layer)
 - Scenario Simulator
-  Readiness (PMRS)
 - City Scorecard
 - Ward Rankings
 - Ward Detail View (per ward)
 - Data Entry Forms
 - Action Items Tracker
 - Report Generator
-  Rainfall
 - Current Conditions

- Forecast View
- Historical Analysis
- IDF Curves
- Station Management
-  Alerts & Operations (Monsoon Mode)
 - Active Alerts Dashboard
 - Deployment Plan
 - Field Team Tracker
 - Citizen Reports Feed
 - Event Timeline
-  Analytics & Reports
 - Trend Analysis
 - Performance Metrics
 - Report Library
 - Data Export
-  Administration
 - User Management
 - Configuration
 - Data Management
 - System Health
 - Audit Logs
-  Help & Documentation
 - User Guide
 - Video Tutorials
 - Glossary
 - Contact Support

6.2 Key Screen Wireframes

text



- 10:15 Red Alert Ward 12
85mm/hr, 12 hotspots
- ☑ 9:30 Orange Alert Ward 31
60mm/hr, 5 hotspots
- 08:45 Yellow Alert Ward 45
35mm/hr, 2 hotspots

- Desilting Ward 08 Due:3d
 - Pump repair W31 Due:5d
 - Barrier test W12 Done
 - Team drill W45 Due:7d
- [View All Actions →]

SCREEN: MAP EXPLORER

UFHE

Dashboard [Map] Hotspots Readiness Rainfall Alerts ⚙

LAYERS

- ☑ Wards
- ☑ Hotspots
- Micro-basins
- ☑ Drainage
- DEM
- Land Use
- Buildings
- Water Bodies
- Flood History

BASEMAP

- Street
- Satellite
- Terrain

OPACITY

70%

Search location...



← Tools

INTERACTIVE MAP

(Mapbox GL / Deck.gl)

LEGEND

- Very High (FVI >0.8)
- High (0.6-0.8)
- Moderate (0.4-0.6)
- Low (0.2-0.4)
- Very Low (<0.2)

FEATURE INFO

Selected:
Micro-Basin
#MB-1247

Area: 0.12 km²

Avg Elev: 42m

Slope: 1.2°

CN: 87

FVI: 0.73 HIGH

Drain Cap:

0.8 m³/s

Peak Runoff:

1.4 m³/s

DEFICIT: 43%

Hist. Floods:

8 events

(last 5 yrs)

[Run Scenario]

[View History]

[Export Data]

SCREEN: WARD READINESS DETAIL (PMRS)

← Back to Rankings

WARD 12: Koramangala

PMRS: 42/100

Grade: C

SPIDER CHART

Infrastructure
35/100

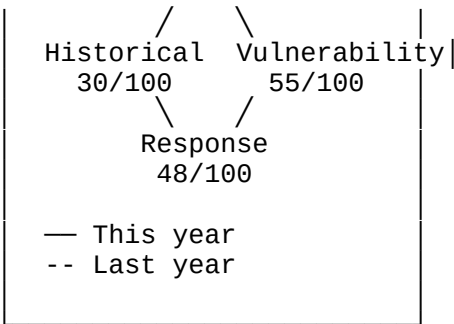
COMPONENT BREAKDOWN

Infrastructure (30%)



35





Desilting: 20% complete
Pumps: 1/4 operational
Drain repairs: 3/8 done
New infra: None planned

Vulnerability (25%)

55

Hotspots covered: 8/15
Population exposed: 12,400
Critical facilities: 2 unprotected
Trend: Worsening (▼ 5 pts)

Response Preparedness (25%)

48

Equipment: 60% positioned
Teams: 3/5 trained
Comms drill: Not done
Evac routes: 2/3 cleared

Historical Performance (20%)

30

Last monsoon: 23 flood events
Avg response: 5.2 hours
Complaints resolved: 45%
Prediction accuracy: 62%

 ACTION ITEMS TO IMPROVE PMRS (Auto-Generated, Priority Order)

PRIO	ACTION	IMPACT	DUE	STATUS
1	Complete desilting of Main Drain (3.2 km left)	+12 pts	Jun 05	 Behind schedule
2	Repair Pump #2 and #4 at Station B	+8 pts	Jun 01	 In progress
3	Conduct communication drill for flood response	+5 pts	Jun 10	 Not started
4	Clear debris at Junction X (critical hotspot)	+4 pts	Jun 03	 Not started
5	Install flood barrier at underpass entry	+3 pts	Jun 15	 Procuring

 Generate Ward Report  Email to Ward Engineer  Print

7. API SPECIFICATION (Key Endpoints)
text

API DESIGN (RESTful)		
Base URL: /api/v1		
SPATIAL DATA		
GET	/wards	List all wards
GET	/wards/{ward_id}	Ward detail + geometry
GET	/wards/{ward_id}/hotspots	Hotspots in a ward
GET	/wards/{ward_id}/drainage	Drainage network in ward
GET	/wards/{ward_id}/micro-basins	Micro-basins in ward
GET	/micro-basins	All micro-basins (paginated)
GET	/micro-basins/{basin_id}	Basin detail + attributes

GET	/micro-basins/{basin_id}/fvi	FVI breakdown
GET	/hotspots	All hotspots (filterable)
GET	/hotspots/{hotspot_id}	Hotspot detail
GET	/hotspots/within ?geometry=<GeoJSON>	Hotspots within polygon (spatial query)
GET	/hotspots/nearby ?lat=X&lng=Y&radius=500m	Hotspots near point
PMRS		
GET	/pmrs/scores	All ward PMRS scores
GET	/pmrs/scores/{ward_id}	Ward PMRS detail
GET	/pmrs/scores/{ward_id}/history	PMRS over time
POST	/pmrs/data-entry	Submit readiness data
GET	/pmrs/action-items/{ward_id}	Action items for ward
PATCH	/pmrs/action-items/{item_id}	Update action item status
GET	/pmrs/report/{ward_id}	Generate ward report
GET	/pmrs/report/city	Generate city report
RAINFALL		
GET	/rainfall/current	Current readings (all stn)
GET	/rainfall/current/{ward_id}	Current for a ward
GET	/rainfall/forecast	Latest forecast
GET	/rainfall/historical ?start=YYYY-MM-DD&end=YYYY-MM-DD	Historical data
GET	/rainfall/idf	IDF curve data
GET	/rainfall/accumulation ?duration=1hr 3hr 6hr 24hr	Running accumulation
PREDICTIONS		
POST	/predict/scenario Body: {rainfall_mm, duration_hr, return_period}	Run flood scenario
GET	/predict/realtime	Current predictions
GET	/predict/realtime/{ward_id}	Predictions for ward
GET	/predict/accuracy	Prediction accuracy stats
ALERTS		
GET	/alerts/active	Current active alerts
GET	/alerts/history	Past alerts
POST	/alerts/acknowledge/{alert_id}	Acknowledge an alert
RESOURCES		
GET	/resources/inventory	All resources
GET	/resources/deployed	Currently deployed
POST	/resources/deploy	Deploy resource
GET	/resources/deployment-plan	Suggested deployment
CITIZEN REPORTS		
POST	/reports Body: {location, photo, description, severity}	Submit flood report
GET	/reports	List reports (filterable)
PATCH	/reports/{report_id}	Update report status
EXPORT		
GET	/export/hotspots ?format=csv shapefile geojson kml	Export as CSV/SHP/GeoJSON
GET	/export/pmrs	Export PMRS data

```

GET      /export/rainfall      Export rainfall data

RESPONSE FORMAT (Standard):
{
  "status": "success",
  "data": { ... },
  "meta": {
    "total": 2847,
    "page": 1,
    "per_page": 50,
    "timestamp": "2024-07-15T10:30:00Z"
  }
}

SPATIAL RESPONSE (GeoJSON):
{
  "type": "FeatureCollection",
  "features": [
    {
      "type": "Feature",
      "geometry": { "type": "Point", "coordinates": [77.6, 12.9] },
      "properties": {
        "hotspot_id": "HS-1247",
        "severity": "HIGH",
        "predicted_depth_cm": 65,
        "fvi_score": 0.73
      }
    }
  ]
}

```

8. DATABASE SCHEMA (Key Tables)

SQL

```

-- =====
-- UFHE DATABASE SCHEMA (PostgreSQL + PostGIS)
-- =====

-- =====
-- SPATIAL CORE TABLES
-- =====

CREATE TABLE wards (
  ward_id      SERIAL PRIMARY KEY,
  ward_name    VARCHAR(100) NOT NULL,
  ward_code    VARCHAR(20) UNIQUE NOT NULL,
  zone_name    VARCHAR(100),
  area_km2     NUMERIC(10,4),
  population   INTEGER,
  geometry     GEOMETRY(MultiPolygon, 4326) NOT NULL,
  created_at   TIMESTAMPTZ DEFAULT NOW(),
  updated_at   TIMESTAMPTZ DEFAULT NOW()
);
CREATE INDEX idx_wards_geom ON wards USING GIST(geometry);

CREATE TABLE micro_basins (
  basin_id     SERIAL PRIMARY KEY,
  basin_code   VARCHAR(20) UNIQUE NOT NULL,
  ward_id      INTEGER REFERENCES wards(ward_id),
  area_km2     NUMERIC(10,6),
  mean_elevation NUMERIC(8,2),      -- meters
  min_elevation NUMERIC(8,2),
  max_elevation NUMERIC(8,2),
  mean_slope   NUMERIC(6,3),      -- degrees

```



```

    pour_point          GEOMETRY(Point, 4326),
    imperviousness      NUMERIC(5,2),      -- percentage
    curve_number        NUMERIC(5,2),
    twi                 NUMERIC(8,3),      -- topographic wetness index
    upstream_area       NUMERIC(10,4),     -- km2
    fvi_score           NUMERIC(4,3),      -- 0.000 to 1.000
    fvi_class           VARCHAR(20),       -- VERY_HIGH, HIGH, MODERATE, LOW, VERY_LOW
    geometry            GEOMETRY(MultiPolygon, 4326) NOT NULL,
    created_at          TIMESTAMPTZ DEFAULT NOW(),
    updated_at          TIMESTAMPTZ DEFAULT NOW()
);
CREATE INDEX idx_basins_geom ON micro_basins USING GIST(geometry);
CREATE INDEX idx_basins_ward ON micro_basins(ward_id);
CREATE INDEX idx_basins_fvi ON micro_basins(fvi_score DESC);

CREATE TABLE drainage_network (
    drain_id            SERIAL PRIMARY KEY,
    drain_code          VARCHAR(30),
    ward_id             INTEGER REFERENCES wards(ward_id),
    drain_type          VARCHAR(30),       -- PRIMARY, SECONDARY, TERTIARY, NALA
    material            VARCHAR(30),       -- CONCRETE, BRICK, RCC, EARTH
    shape              VARCHAR(20),       -- CIRCULAR, RECTANGULAR, TRAPEZOIDAL
    width_m             NUMERIC(6,3),
    height_m           NUMERIC(6,3),
    diameter_m          NUMERIC(6,3),     -- for circular pipes
    length_m            NUMERIC(10,2),
    slope              NUMERIC(8,6),      -- m/m
    manning_n           NUMERIC(5,4),
    capacity_m3s        NUMERIC(8,4),     -- computed full capacity
    eff_capacity        NUMERIC(8,4),     -- effective after degradation
    age_years           INTEGER,
    condition           VARCHAR(20),      -- GOOD, FAIR, POOR, CRITICAL
    last_desilt         DATE,
    geometry            GEOMETRY(MultiLineString, 4326) NOT NULL,
    created_at          TIMESTAMPTZ DEFAULT NOW()
);
CREATE INDEX idx_drain_geom ON drainage_network USING GIST(geometry);

CREATE TABLE flood_hotspots (
    hotspot_id          SERIAL PRIMARY KEY,
    hotspot_code        VARCHAR(20) UNIQUE NOT NULL,
    basin_id            INTEGER REFERENCES micro_basins(basin_id),
    ward_id             INTEGER REFERENCES wards(ward_id),
    location_name       VARCHAR(200),
    fvi_score           NUMERIC(4,3),
    severity_class      VARCHAR(20),
    predicted_depth     NUMERIC(6,2),     -- cm, for design storm
    drain_deficit       NUMERIC(5,2),     -- percentage
    pop_exposed         INTEGER,
    critical_fac        INTEGER,          -- nearby critical facilities count
    latitude            NUMERIC(10,7),
    longitude           NUMERIC(10,7),
    geometry            GEOMETRY(Point, 4326) NOT NULL,
    rank_city           INTEGER,
    rank_ward           INTEGER,
    created_at          TIMESTAMPTZ DEFAULT NOW(),
    updated_at          TIMESTAMPTZ DEFAULT NOW()
);
CREATE INDEX idx_hotspot_geom ON flood_hotspots USING GIST(geometry);
CREATE INDEX idx_hotspot_severity ON flood_hotspots(severity_class);

```

```

-- -----
-- RAINFALL TABLES
-- -----

```

```

CREATE TABLE rain_stations (
    station_id      SERIAL PRIMARY KEY,
    station_code    VARCHAR(20) UNIQUE,
    station_name     VARCHAR(100),
    station_type     VARCHAR(20),          -- AWS, MANUAL, RADAR
    ward_id         INTEGER REFERENCES wards(ward_id),
    latitude         NUMERIC(10,7),
    longitude        NUMERIC(10,7),
    elevation_m      NUMERIC(8,2),
    status           VARCHAR(20),          -- ACTIVE, INACTIVE, MAINTENANCE
    geometry         GEOMETRY(Point, 4326)
);

CREATE TABLE rainfall_observations (
    obs_id          BIGSERIAL PRIMARY KEY,
    station_id      INTEGER REFERENCES rain_stations(station_id),
    timestamp       TIMESTAMPTZ NOT NULL,
    rainfall_mm     NUMERIC(8,2),          -- cumulative for interval
    interval_min    INTEGER,              -- 15, 30, 60 minutes
    quality_flag    VARCHAR(10),          -- GOOD, SUSPECT, MISSING
    source          VARCHAR(20)           -- AWS, MANUAL, SATELLITE
);

CREATE INDEX idx_rain_obs_time ON rainfall_observations(timestamp);
CREATE INDEX idx_rain_obs_station ON rainfall_observations(station_id);
-- Partitioning by month recommended for large datasets

CREATE TABLE rainfall_basin (
    id              BIGSERIAL PRIMARY KEY,
    basin_id        INTEGER REFERENCES micro_basins(basin_id),
    timestamp       TIMESTAMPTZ NOT NULL,
    rainfall_mm     NUMERIC(8,2),          -- interpolated value
    interp_method   VARCHAR(20),          -- KRIGING, IDW, THIESSEN
    accumulation_1h NUMERIC(8,2),
    accumulation_3h NUMERIC(8,2),
    accumulation_6h NUMERIC(8,2),
    accumulation_24h NUMERIC(8,2)
);

-- -----
-- PMRS TABLES
-- -----

CREATE TABLE pmrs_scores (
    score_id        SERIAL PRIMARY KEY,
    ward_id         INTEGER REFERENCES wards(ward_id),
    assessment_year  INTEGER NOT NULL,
    assessment_date  DATE,

    -- Component scores (0-100)
    infra_score     NUMERIC(5,2),
    vuln_score      NUMERIC(5,2),
    response_score  NUMERIC(5,2),
    history_score   NUMERIC(5,2),

    -- Sub-component scores
    desilting_pct  NUMERIC(5,2),
    pumps_operational NUMERIC(5,2),
    repairs_done   NUMERIC(5,2),
    hotspot_coverage NUMERIC(5,2),

    -- Final score
    total_score     NUMERIC(5,2),
    grade           CHAR(1),              -- A, B, C, D, F

```

```

        computed_at      TIMESTAMPTZ DEFAULT NOW(),

        UNIQUE(ward_id, assessment_year)
    );

CREATE TABLE pmrs_action_items (
    item_id              SERIAL PRIMARY KEY,
    ward_id              INTEGER REFERENCES wards(ward_id),
    assessment_year      INTEGER,
    priority              INTEGER,
    description           TEXT,
    impact_points         NUMERIC(5,2),      -- estimated PMRS improvement
    due_date              DATE,
    assigned_to           INTEGER REFERENCES users(user_id),
    status                VARCHAR(20),       -- NOT_STARTED, IN_PROGRESS, DONE, OVERDUE
    completion_date       DATE,
    notes                 TEXT,
    created_at            TIMESTAMPTZ DEFAULT NOW()
);

-- -----
-- PREDICTION & EVENTS TABLES
-- -----

CREATE TABLE flood_predictions (
    prediction_id         BIGSERIAL PRIMARY KEY,
    basin_id              INTEGER REFERENCES micro_basins(basin_id),
    hotspot_id            INTEGER REFERENCES flood_hotspots(hotspot_id),
    prediction_time        TIMESTAMPTZ,      -- when prediction was made
    valid_time             TIMESTAMPTZ,      -- what time period it's for
    rainfall_input         NUMERIC(8,2),     -- mm used as input
    predicted_depth        NUMERIC(6,2),     -- cm
    predicted_volume       NUMERIC(12,2),    -- m³
    confidence             VARCHAR(10),      -- HIGH, MEDIUM, LOW
    model_version          VARCHAR(20),
    alert_level            VARCHAR(10),      -- GREEN, YELLOW, ORANGE, RED
    actual_depth           NUMERIC(6,2),     -- filled in post-event (for validation)
    created_at             TIMESTAMPTZ DEFAULT NOW()
);

CREATE TABLE flood_events (
    event_id              SERIAL PRIMARY KEY,
    event_date             DATE,
    event_name             VARCHAR(100),     -- "July 15 2024 Heavy Rainfall"
    total_rainfall         NUMERIC(8,2),     -- city average mm
    peak_intensity         NUMERIC(8,2),     -- mm/hr
    duration_hours         NUMERIC(6,2),
    wards_affected         INTEGER,
    hotspots_activated     INTEGER,
    total_complaints       INTEGER,
    notes                  TEXT
);

CREATE TABLE flood_event_observations (
    obs_id                BIGSERIAL PRIMARY KEY,
    event_id              INTEGER REFERENCES flood_events(event_id),
    hotspot_id            INTEGER REFERENCES flood_hotspots(hotspot_id),
    basin_id              INTEGER REFERENCES micro_basins(basin_id),
    observed_depth         NUMERIC(6,2),     -- cm
    observation_time       TIMESTAMPTZ,
    duration_hours         NUMERIC(6,2),     -- how long waterlogging lasted
    source                 VARCHAR(30),      -- FIELD_SURVEY, CITIZEN_REPORT, SENSOR,
    SATELLITE

```

```

        photo_url      TEXT,
        verified       BOOLEAN DEFAULT FALSE
    );

-- -----
-- USER & SYSTEM TABLES
-- -----

CREATE TABLE users (
    user_id            SERIAL PRIMARY KEY,
    username           VARCHAR(50) UNIQUE NOT NULL,
    email              VARCHAR(100) UNIQUE,
    phone              VARCHAR(15),
    full_name          VARCHAR(100),
    role               VARCHAR(30),          -- SUPER_ADMIN, CITY_ADMIN, ZONAL_OFFICER,
                                           -- WARD_ENGINEER, FIELD_STAFF, VIEWER,
    ward_id            INTEGER REFERENCES wards(ward_id), -- NULL for city-level
    zone              VARCHAR(50),
    is_active          BOOLEAN DEFAULT TRUE,
    password_hash      VARCHAR(255),
    last_login         TIMESTAMPTZ,
    created_at         TIMESTAMPTZ DEFAULT NOW()
);

CREATE TABLE audit_log (
    log_id             BIGSERIAL PRIMARY KEY,
    user_id            INTEGER REFERENCES users(user_id),
    action             VARCHAR(50),          -- CREATE, UPDATE, DELETE, LOGIN, EXPORT
    entity_type        VARCHAR(50),          -- WARD, HOTSPOT, PMRS, PREDICTION
    entity_id          INTEGER,
    details            JSONB,
    ip_address         INET,
    timestamp          TIMESTAMPTZ DEFAULT NOW()
);

CREATE TABLE citizen_reports (
    report_id          SERIAL PRIMARY KEY,
    reporter_phone     VARCHAR(15),
    reporter_name      VARCHAR(100),
    location           GEOMETRY(Point, 4326),
    ward_id            INTEGER REFERENCES wards(ward_id),
    nearest_hotspot    INTEGER REFERENCES flood_hotspots(hotspot_id),
    description        TEXT,
    severity_reported  VARCHAR(20),
    photo_urls         TEXT[],
    status             VARCHAR(20),          -- SUBMITTED, VERIFIED, ASSIGNED, RESOLVED
    assigned_to        INTEGER REFERENCES users(user_id),
    resolution_time    INTERVAL,
    submitted_at       TIMESTAMPTZ DEFAULT NOW(),
    resolved_at        TIMESTAMPTZ
);
9. RELEASE PLAN
text

```

RELEASE PLAN
RELEASE 1.0 – "FOUNDATION" (Month 4) ✓ GIS map with ward boundaries, drainage, DEM ✓ Micro-basin delineation (2,500+) ✓ FVI computation and hotspot identification

- ✓ Static scenario simulation (design storms)
 - ✓ Basic user auth and roles
 - ✓ Data export (CSV, GeoJSON)
- TARGET USERS: GIS team, hydrology consultants

RELEASE 2.0 – "READINESS" (Month 7)

- ✓ PMRS scoring engine
 - ✓ Data entry forms for ward engineers
 - ✓ PMRS dashboard with rankings & spider charts
 - ✓ Action item generation and tracking
 - ✓ Report generation (PDF)
 - ✓ Historical rainfall analysis & IDF curves
- TARGET USERS: Ward engineers, zonal officers, commissioner

RELEASE 3.0 – "PREDICTION" (Month 10)

- ✓ Real-time rainfall ingestion
 - ✓ ML-enhanced flood predictions
 - ✓ Alert system (in-app, SMS, email)
 - ✓ Resource deployment suggestions
 - ✓ Temporal flood progression view
- TARGET USERS: All internal users, control room

RELEASE 4.0 – "OPERATIONS" (Month 13)

- ✓ Mobile app for field teams
 - ✓ Citizen reporting module
 - ✓ Real-time deployment tracking
 - ✓ Post-event analysis tools
 - ✓ What-if infrastructure simulator
- TARGET USERS: Field teams, citizens, full organization

RELEASE 5.0+ – "INTELLIGENCE" (Month 16+)

- ✓ 3D visualization
- ✓ Social media NLP integration
- ✓ Climate change projections
- ✓ Economic impact assessment
- ✓ IoT sensor integration
- ✓ Multi-city support

PART B: COMPLETE DATA SOURCE DIRECTORY

Where to Find Every Piece of Data You Need

1. DIGITAL ELEVATION MODEL (DEM) SOURCES

text

DEM / TERRAIN DATA

★ FREE GLOBAL SOURCES

1. SRTM (Shuttle Radar Topography Mission)

Resolution: 30m (1 arc-second)

Coverage: Global (60°N to 56°S)

Download:

→ <https://earthexplorer.usgs.gov>

(Register → Search → Dataset: SRTM 1 Arc-Second Global)

→ <https://dwtkns.com/srtm30m/>

(Direct tile download, easier interface)

→ <https://opentopography.org>

(Bulk download with API)

Format: GeoTIFF (.tif)
QUALITY: OK for regional analysis, TOO COARSE for urban micro-basins. Use as fallback only.

2. ALOS PALSAR DEM (JAXA - Japan)

Resolution: 12.5m

Coverage: Global

Download:

→ <https://search.asf.alaska.edu>

(Alaska Satellite Facility – free registration)

Dataset: ALOS PALSAR → Radiometric Terrain Corrected

→ https://www.eorc.jaxa.jp/ALOS/en/dataset/aw3d30/aw3d30_e.htm

(ALOS World 3D – 30m, also free)

Format: GeoTIFF

QUALITY: GOOD balance of resolution and coverage.

★ RECOMMENDED AS PRIMARY FREE SOURCE

3. FABDEM (Forest And Buildings removed Copernicus DEM)

Resolution: 30m

Coverage: Global

Download:

→ <https://data.bris.ac.uk/data/dataset/s5hqmjcdj8yo2ibzi9b4ew3sn>

SPECIAL: Buildings and trees REMOVED = closer to true ground elevation. VERY USEFUL for urban flood modeling.

Format: GeoTIFF

QUALITY: ★★★★★ Best free option for urban hydrology

4. Copernicus DEM (European Space Agency)

Resolution: 30m (GLO-30) and 90m (GLO-90)

Download:

→ <https://spacedata.copernicus.eu/collections/copernicus-digital-elevation-model>

→ Via OpenTopography API

Format: GeoTIFF

QUALITY: Very accurate, recent (2021)

5. CartoDEM (ISRO – India specific) ★★★★★

Resolution: ~30m (derived from Cartosat stereo imagery)

Coverage: India

Download:

→ <https://bhuvan.nrsc.gov.in>

(Register → Bhuvan → Download → CartoDEM)

→ <https://bhuvan-app1.nrsc.gov.in/thematic/thematic/index.php>

Format: GeoTIFF

QUALITY: BEST FREE SOURCE FOR INDIAN CITIES.

Specifically designed for India's terrain.

★ PAID / HIGH-RESOLUTION SOURCES

6. LiDAR Survey (Custom Commission)

Resolution: 0.5m - 2m

How to get:

→ Contact survey companies: Geokno, ScanPoint, Trimble India

→ Municipal corporation may already have (check with Survey of India or city planning department)

→ Cost: ₹5-15 per hectare (airborne LiDAR)

→ National LiDAR Mapping Program (NLMP) by Survey of India may have coverage – check with SOI

QUALITY: ★★★★★ GOLD STANDARD. Get this for top 50 hotspots.

7. Drone Photogrammetry (DIY High-Res DEM)

Resolution: 2cm - 50cm

How to do:

- DJI Phantom/Mavic drone with survey-grade flight planning
- Software: Pix4D, Agisoft Metashape, OpenDroneMap (free)
- Fly critical flood-prone areas (underpasses, low-lying zones)
- Cost: ~₹2-5L for equipment + ₹5K per km² for processing
- DGCA approval needed for urban drone flights in India
- QUALITY: ★★★★★ Best for specific critical locations

8. Maxar / Airbus WorldDEM

- Resolution: 5m
- <https://www.maxar.com>
- <https://www.intelligence-airbusds.com/worldDEM/>
- Cost: Commercial pricing (thousands of \$)
- QUALITY: Excellent but expensive

★ RECOMMENDED STRATEGY:

- Layer 1: FABDEM or CartoDEM (30m) for entire city → FREE
- Layer 2: ALOS PALSAR (12.5m) for entire city → FREE
- Layer 3: LiDAR or Drone for top 100 critical hotspots → PAID
- Merge using weighted averaging in overlapping areas.

2. RAINFALL DATA SOURCES

text

RAINFALL DATA

★ INDIAN METEOROLOGICAL DEPARTMENT (IMD) – PRIMARY SOURCE

1. IMD Gridded Rainfall Data (Daily)
 - Resolution: 0.25° × 0.25° (~25km grid)
 - Period: 1901 to present
 - Download:
 - https://www.imdpune.gov.in/cmpg/Griddata/Rainfall_25_Bin.html
 - Contact: cdsp@imdpune.gov.in
 - Data available in binary grid format
 - Python library to read: `imdlib` (pip install imdlib)
 - Format: Binary grid / NetCDF
 - COST: Free for research, nominal fee for commercial
 - USE: Long-term trend analysis, IDF curves, return periods
2. IMD Station-Level Data (Daily/Hourly)
 - <https://cdsp.imdpune.gov.in/> (Climate Data Services Portal)
 - Submit formal request for station data
 - Specify: city name, station codes, date range
 - Hourly data available for AWS stations (post-2000)
 - Processing time: 2-4 weeks
 - COST: ₹500-5000 depending on volume
 - USE: IDF curves, hourly storm patterns
3. IMD AWS (Automatic Weather Stations) – Real-time
 - <http://aws.imd.gov.in>
 - Some cities have 15-minute interval data
 - API access: contact IMD's IT division
 - Check if your city corporation has its own AWS network
 - USE: Real-time monitoring, nowcasting input
4. IMD Radar Data
 - https://mausam.imd.gov.in/imd_latest/contents/radar.php
 - DWR (Doppler Weather Radar) images available
 - Covers most major Indian cities

→ API/bulk access: formal request to IMD
USE: Spatial rainfall estimation, nowcasting

★ SATELLITE-BASED RAINFALL ESTIMATES

5. NASA GPM IMERG (Global Precipitation Measurement)

Resolution: $0.1^\circ \times 0.1^\circ$ (~10km), 30-minute intervals

Latency: Near-real-time (4 hours), Final (3 months)

Download:

→ https://disc.gsfc.nasa.gov/datasets/GPM_3IMERGHH_06/summary

→ <https://pmm.nasa.gov/data-access/downloads/gpm>

→ API: <https://disc.gsfc.nasa.gov/information/tools>

→ Python: Use `requests` with Earthdata Login credentials

Registration: <https://urs.earthdata.nasa.gov/>

Format: HDF5 / NetCDF4

COST: FREE

USE: Gap-fill where ground stations don't exist,
cross-validation with ground truth

★ EXCELLENT for cities with sparse rain gauge network

6. CHIRPS (Climate Hazards Infrared Precipitation)

Resolution: 0.05° (~5km), Daily

Period: 1981 to present

Download:

→ <https://www.chc.ucsb.edu/data/chirps>

→ <https://data.chc.ucsb.edu/products/CHIRPS-2.0/>

Format: GeoTIFF, NetCDF

COST: FREE

USE: Long-term climatology, trend analysis

7. ERA5 Reanalysis (ECMWF)

Resolution: 0.25° (~25km), Hourly

Period: 1950 to present

Download:

→ <https://cds.climate.copernicus.eu/cdsapp#!/dataset/reanalysis-era5-single-levels>

→ Python API: `cdsapi` (pip install cdsapi)

Format: GRIB / NetCDF

COST: FREE

USE: Filling gaps in ground observations, long-term analysis,
also provides soil moisture, temperature, humidity

8. INSAT-3D/3DR (ISRO Satellite)

→ <https://mosdac.gov.in>

(Meteorological & Oceanographic Satellite Data Archival Centre)

→ Real-time and archived satellite imagery

→ Derived rainfall products (IMSRA – merged satellite product)

→ Registration required (free)

Format: HDF5, GeoTIFF

COST: FREE

USE: Real-time cloud imagery, satellite-derived rainfall

★ CITY-SPECIFIC SOURCES

9. Municipal Corporation's Own Rain Gauges

→ Contact: Storm Water Drainage Department

→ Many cities (Mumbai, Chennai, Bengaluru, Hyderabad) have
dense AWS networks installed post-2015

→ Data may be on internal servers / Excel sheets

→ THIS IS YOUR MOST VALUABLE DATA SOURCE

→ Mumbai: MCGM has 60+ rain gauges

→ Chennai: GCC has 25+ AWS stations

→ Bengaluru: BBMP + KSNDMC have 70+ stations

10. State Disaster Management Authority

- Most states maintain rainfall monitoring networks
- KSNDMC (Karnataka): <https://www.ksndmc.org>
- TSDPS (Telangana): <https://tsdps.telangana.gov.in>
- EMSC (Maharashtra): real-time rainfall data
- TNSDMA (Tamil Nadu): <https://tnsdma.tn.gov.in>

11. Central Water Commission (CWC)

- <https://cwc.gov.in>
- River gauge data + rainfall at river stations
- Flood forecasting data
- India-WRIS: <https://indiawris.gov.in/wris/>
- USE: River levels for cities with riverine flooding

12. Weatherunderground / Private Weather Networks

- <https://www.wunderground.com/history>
- Citizen-run weather stations (good for validation)
- API available (limited free tier)

13. Open-Meteo (Free Weather API)

- <https://open-meteo.com/>
- Free API for historical + forecast weather data
- Python: `openmeteo\_requests` library
- Hourly resolution, based on ERA5 + weather models
- Excellent for prototyping before securing IMD data
- ★ BEST FREE API TO START PROTOTYPING IMMEDIATELY

★ REAL-TIME FORECAST SOURCES

14. IMD Forecast (Official)

- <https://mausam.imd.gov.in>
- City-specific forecasts, heavy rain warnings
- QPF (Quantitative Precipitation Forecast)
- Color-coded warnings (Green/Yellow/Orange/Red)

15. GFS (Global Forecast System – NOAA/USA)

- <https://nomads.ncep.noaa.gov>
- Free, updated 4× daily
- Resolution: 0.25°, out to 16 days
- Format: GRIB2
- Good for 3-7 day outlooks

16. ECMWF (European Centre) – Best global model

- <https://www.ecmwf.int/en/forecasts>
- Free for research (registration needed)
- Resolution: 0.1° (HRES), ensemble available
- ★ Most accurate global NWP model

★ PYTHON CODE TO GET STARTED WITH RAINFALL DATA:

```
# Using imdlib for IMD gridded data
pip install imdlib
```

```
import imdlib as imd
# Download rainfall data for a region
start_yr = 2010
end_yr = 2023
variable = 'rain' # 'rain' for rainfall, 'tmax'/'tmin' for temp
data = imd.get_data(variable, start_yr, end_yr, fn_format='yearwise')
ds = imd.open_data(variable, start_yr, end_yr, 'yearwise')
# Get data for specific lat/lon
```

```
ts = ds.get_timeseries([12.97, 77.59]) # Bengaluru coordinates

# Using Open-Meteo API for quick prototyping
import openmeteo_requests
import requests_cache
from retry_requests import retry
cache_session = requests_cache.CachedSession('.cache', expire=3600)
om = openmeteo_requests.Client(session=retry(cache_session))
params = {
    "latitude": 12.97,
    "longitude": 77.59,
    "hourly": "precipitation",
    "start_date": "2020-01-01",
    "end_date": "2023-12-31"
}
responses = om.weather_api(
    "https://archive-api.open-meteo.com/v1/archive", params=params)
```

3. DRAINAGE & INFRASTRUCTURE DATA

text

DRAINAGE NETWORK & INFRASTRUCTURE DATA

1. Municipal Corporation SWD Department
 - Storm Water Drainage (SWD) Department GIS data
 - Often digitized under Smart Cities Mission or AMRUT
 - Ask for: drain line shapefile, pipe dimensions database, pump station locations, outfall points
 - If not digitized: request paper maps and digitize manually using QGIS
 - Contact: Executive Engineer, SWD Division
 2. Smart Cities Mission GIS Data
 - <https://smartcities.gov.in>
 - Many Smart City SPVs have GIS cells with digitized infrastructure
 - Command and Control Centers often have drainage data
 - Check with your city's Smart City SPV
 3. AMRUT Data (Atal Mission for Rejuvenation & Urban Transformation)
 - <https://amrut.gov.in>
 - Cities under AMRUT have prepared Service Level Improvement Plans (SLIPs) with drainage data
 - DPRs (Detailed Project Reports) contain pipe diameters, gradients, capacity calculations
 4. OpenStreetMap (OSM) – Free
 - <https://www.openstreetmap.org>
 - Tags: waterway=drain, waterway=canal, waterway=stream
 - Download via:
 - <https://overpass-turbo.eu> (query specific area)
 - <https://download.geofabrik.de/asia/india.html> (Pre-built country/state extracts)
 - Quality: Variable – good in well-mapped cities, may need supplementing
 - Python: `osmnx` library for easy download
- ```
import osmnx as ox
Get waterways/drainage for a city
G = ox.graph_from_place("Bengaluru, India",
```

```
custom_filter='["waterway"~"drain|canal|stream"]')
```

5. Survey of India Toposheets

- 1:25,000 and 1:50,000 scale
- Show natural drainage lines, contours, water bodies
- Available at SOI offices or online:
  - <https://onlinemaps.surveyofindia.gov.in/FreeOtherMaps.aspx>
- Useful for natural drainage patterns

6. City Master Plan / CDP Documents

- City Development Plans and Master Plans contain:  
drainage network maps, flood-prone zone identification,  
infrastructure capacity assessments
- Usually available as PDF on city corporation website
- Georeferencing PDF maps in QGIS:  
Open PDF → Add ground control points → Export as GeoTIFF  
→ Digitize features manually

4. LAND USE & IMPERVIOUS SURFACE DATA  
text

LAND USE / LAND COVER (LULC) & IMPERVIOUS SURFACE

1. ISRO Bhuvan LULC (India-specific) ★★★★★

- <https://bhuvan.nrsc.gov.in>
- LULC maps at 1:50,000 scale for entire India
- Decadal LULC (2005-06, 2011-12, 2015-16)
- Download: Register → Thematic Data → LULC
- Format: Shapefile / GeoTIFF
- COST: FREE

2. ESA WorldCover (European Space Agency)

- <https://esa-worldcover.org/en>
- Resolution: 10m (Sentinel-2 derived)
- 11 land cover classes including "Built-up"
- Latest: 2021
- Download: direct GeoTIFF tiles
- COST: FREE
- ★ BEST FREE HIGH-RESOLUTION GLOBAL LULC

3. Google Dynamic World

- <https://dynamicworld.app>
- Resolution: 10m, near-real-time updates
- 9 land cover classes
- Access via Google Earth Engine
- Python: `ee` library (Google Earth Engine API)
- COST: FREE

4. Global Impervious Surface (For Runoff Calculation)

- GAIA dataset: <https://doi.org/10.1594/PANGAEA.897975>  
(30m global impervious area, 1985-2018)
- Global Urban Footprint (DLR):  
<https://www.dlr.de/eoc/en/desktopdefault.aspx/tabid-11725/>
- GHSL (Global Human Settlement Layer):  
<https://ghsl.jrc.ec.europa.eu/download.php>  
Built-up surface, population grids
- COST: FREE

5. NDVI for Green Cover (Derive from Satellite)

- Sentinel-2 imagery via Copernicus Open Access Hub  
<https://scihub.copernicus.eu>
- Google Earth Engine (easiest)

- Calculate NDVI = (NIR - Red) / (NIR + Red)
  - High NDVI = vegetation = higher infiltration
  - Low NDVI = built-up/impervious = more runoff
- COST: FREE

#### 6. Building Footprints

- Microsoft Building Footprints:  
<https://github.com/microsoft/GlobalMLBuildingFootprints>  
 (ML-extracted, covers India, FREE)
  - Google Open Buildings:  
<https://sites.research.google/open-buildings/>  
 (670M+ buildings, covers India, FREE)
  - OpenStreetMap buildings (via Overpass API)
- USE: Building density, impervious area estimation

### 5. SOIL & HYDROGEOLOGY DATA

text

#### SOIL & GROUNDWATER DATA

##### 1. NBSS&LUP Soil Maps (India)

- National Bureau of Soil Survey & Land Use Planning
- <https://www.nbsslup.in>
- Soil type maps at district level
- Needed for: Hydrological Soil Group (A/B/C/D) for CN method
- May need formal request; some data on Bhuvan

##### 2. FAO Harmonized World Soil Database

- <https://www.fao.org/soils-portal/data-hub/>
  - Resolution: ~1km
  - Soil texture, drainage class, water holding capacity
- COST: FREE

##### 3. SoilGrids (ISRIC)

- <https://soilgrids.org>
  - Resolution: 250m
  - Clay content, sand content, organic carbon
  - Can derive Hydrological Soil Group from texture
  - API + bulk download available
- COST: FREE

★ BEST FREE SOIL DATA FOR THIS PROJECT

##### 4. CGWB (Central Ground Water Board)

- <http://cgwb.gov.in>
- Water table depth maps
- Shallow water table = less infiltration capacity
- District-level reports available as PDF
- India-WRIS also has groundwater data

#### HYDROLOGICAL SOIL GROUP MAPPING:

| Group      | Texture          | Infiltration Rate |
|------------|------------------|-------------------|
| A (High)   | Sand, gravel     | > 7.6 mm/hr       |
| B (Mod.)   | Sandy loam       | 3.8 - 7.6 mm/hr   |
| C (Slow)   | Clay loam        | 1.3 - 3.8 mm/hr   |
| D (V.Slow) | Clay, silty clay | < 1.3 mm/hr       |

### 6. HISTORICAL FLOOD DATA

text

## HISTORICAL FLOOD RECORDS & VALIDATION DATA

1. Municipal Corporation Complaint Data
  - Waterlogging/flooding complaints from citizen portals
  - Examples:
    - Mumbai: <https://portal.mcgm.gov.in>
    - Bengaluru: BBMP Sahaaya app / Janaagraha
    - Chennai: GCC grievance portal
    - Hyderabad: GHMC complaint system
  - Get geotagged complaints with timestamps
  - CRITICAL for training ML model
2. NDMA / SDMA Flood Reports
  - <https://ndma.gov.in>
  - State Disaster Management Authority reports
  - Post-event damage assessments
  - Flood affected area maps
3. Sentinel-1 SAR Flood Maps (Satellite)
  - Copernicus EMS (Emergency Management Service):  
<https://emergency.copernicus.eu/mapping/>
  - Activated flood maps for major events
  - Sentinel-1 SAR imagery (sees through clouds!):  
<https://scihub.copernicus.eu>
  - Derive flood extent: water appears dark in SAR
  - Tools: SNAP (ESA), Google Earth Engine
  - COST: FREE
  - USE: Ground truth for validating model predictions
4. News Archive Mining
  - Search local newspaper archives for flood reports
  - Extract: date, location, approximate depth, duration
  - Sources: Times of India, Hindu, local language papers
  - Can use NLP to semi-automate extraction
5. Social Media Mining
  - Twitter/X API: search for "flood" + city name
  - Geotagged photos and videos
  - Reddit city subreddits
  - Google Photos (shared publicly)
  - YouTube (flood videos with location in description)
  - Tools: snsrape (Python), Tweepy
6. Dartmouth Flood Observatory
  - <https://floodobservatory.colorado.edu>
  - Global Active Archive of Large Flood Events
  - Satellite-derived flood extent maps
  - COST: FREE
7. EM-DAT International Disaster Database
  - <https://www.emdat.be>
  - Major flood events cataloged globally
  - Registration required
  - COST: FREE
8. India-WRIS Flood Atlas
  - <https://indiawris.gov.in/wris/>
  - Flood-prone area maps, flood frequency maps
  - River basin wise flood history
  - COST: FREE

## 7. WARD BOUNDARIES & ADMINISTRATIVE DATA

text

### ADMINISTRATIVE BOUNDARIES & BASE DATA

1. Ward Boundaries
  - Municipal Corporation website (check GIS/maps section)
  - State Election Commission (ward delimitation maps)
  - DataMeet India: <https://github.com/datameet> (Community-maintained admin boundary shapefiles)
  - Bhuvan: Administrative boundaries download
  - Survey of India: village/ward boundaries
2. Census Data (Population per Ward)
  - <https://censusindia.gov.in>
  - Ward-level population (2011 Census, projections for current)
  - Can also use WorldPop: <https://www.worldpop.org> (100m gridded population, updated annually, FREE)
3. Road Network
  - OpenStreetMap (best free source)
  - Download: <https://download.geofabrik.de>
  - Python: `osmnx` library
4. Water Bodies
  - OpenStreetMap (natural=water tag)
  - Bhuvan water body layer
  - Global Surface Water (JRC): <https://global-surface-water.appspot.com>
  - Shows water body extent changes over decades
  - Identifies encroached water bodies (key flood cause!)
5. Critical Facilities (Hospitals, Schools, Substations)
  - OpenStreetMap POI data
  - Google Places API (paid, good coverage)
  - City corporation records
  - National Health Mission facility lists
  - State education department school lists
6. Google Earth / Bing Maps Imagery
  - Use as basemap in QGIS via XYZ Tiles:
    - Google: <https://mt1.google.com/vt/lyrs=s&x={x}&y={y}&z={z}>
    - Bing: <https://ecn.t0.tiles.virtualearth.net/tiles/a{q}.jpeg>
    - ESRI: [https://server.arcgisonline.com/ArcGIS/rest/services/World\\_Imagery/MapServer/tile/{z}/{y}/{x}](https://server.arcgisonline.com/ArcGIS/rest/services/World_Imagery/MapServer/tile/{z}/{y}/{x})
  - Manual digitization of features from satellite imagery
  - Change detection between years

## 8. QUICK-START DATA ACQUISITION PLAN

text

### "I WANT TO START BUILDING THIS WEEK" – MINIMUM DATA KIT

DAY 1: Download these (all FREE, no approvals needed):

- ☐ DEM:
  - FABDEM from University of Bristol
  - OR CartoDEM from Bhuvan

→ Download tiles covering your city

□ City Boundary + Roads + Water Bodies + Buildings:

→ OpenStreetMap via Geofabrik or osmnx

```
pip install osmnx
```

```
import osmnx as ox
```

```
city = ox.geocode_to_gdf("Bengaluru, India")
```

```
roads = ox.graph_from_place("Bengaluru", network_type='drive')
```

```
buildings = ox.features_from_place("Bengaluru",
 tags={'building': True})
```

□ Land Use / Land Cover:

→ ESA WorldCover 10m tiles

→ Download from: <https://esa-worldcover.org>

□ Building Footprints:

→ Google Open Buildings or Microsoft Building Footprints

□ Rainfall (for prototyping):

→ Open-Meteo API (no key needed!)

→ OR: pip install imdlib → download IMD gridded data

□ Soil:

→ SoilGrids 250m from <https://soilgrids.org>

□ Population:

→ WorldPop 100m grid from <https://worldpop.org>

DAY 2-3: Process and load into PostGIS

□ Install: PostgreSQL + PostGIS

→ Docker: `docker run -d -p 5432:5432 postgres/postgis`

□ Load DEM and run watershed delineation:

```
pip install pysheds richdem whitebox
```

```
from pysheds.grid import Grid
grid = Grid.from_raster('dem.tif')
dem = grid.read_raster('dem.tif')
pit_filled = grid.fill_pits(dem)
flooded = grid.fill_depressions(pit_filled)
inflated = grid.resolve_flats(flooded)
fdir = grid.flowdir(inflated)
acc = grid.accumulation(fdir)
Extract streams and basins...
```

□ Load all vector data into PostGIS:

```
ogr2ogr -f "PostgreSQL" PG:"host=localhost dbname=ufhe
user=postgres" ward_boundaries.shp -nln wards
```

WEEK 1: Apply SCS-CN method to micro-basins

□ Compute CN per basin from LULC + Soil

□ Compute drainage capacity from available network data

□ Compute FVI for each basin

□ Generate first hotspot map → VISUALIZE IN QGIS

WEEK 2: Build basic dashboard

□ FastAPI backend serving GeoJSON

□ React + Mapbox GL frontend showing hotspot map

☐ Basic ward PMRS skeleton

★ YOU NOW HAVE A WORKING PROTOTYPE! ★

WEEK 3-4: Request official data

- ☐ Write formal letter to Municipal Corporation SWD Dept.  
requesting: drainage network GIS, pump station data,  
desilting records, complaint data
- ☐ Request IMD station-level hourly data
- ☐ Request ward boundary shapefile from corporation
- ☐ Collect historical flood records from newspaper archives

These take 2-6 weeks to get responses. While waiting,  
you have enough free data to build 80% of the system.

## 9. DATA TOOLS & LIBRARIES CHEATSHEET

text

### ESSENTIAL PYTHON LIBRARIES FOR THIS PROJECT

```
SPATIAL / GIS
pip install geopandas # Vector data handling
pip install rasterio # Raster data handling
pip install shapely # Geometric operations
pip install fiona # File I/O for spatial data
pip install pyproj # Coordinate transformations
pip install folium # Quick interactive maps
pip install contextily # Basemap tiles for matplotlib

HYDROLOGY SPECIFIC
pip install pysheds # DEM-based watershed delineation
pip install richdem # Terrain analysis
pip install whitebox # WhiteboxTools (comprehensive)
pip install pyflwdir # Flow direction algorithms

RAINFALL / CLIMATE
pip install imdlib # IMD gridded data reader
pip install netCDF4 # NetCDF file handling
pip install xarray # N-dimensional data arrays
pip install cfrib # GRIB file handling
pip install openmeteo-requests # Open-Meteo weather API

DATA & ML
pip install pandas numpy scipy
pip install scikit-learn
pip install xgboost lightgbm
pip install shap # Model explainability
pip install mlflow # Experiment tracking

DATABASE
pip install sqlalchemy # ORM
pip install geoalchemy2 # PostGIS integration
pip install psycopg2-binary # PostgreSQL driver

API & WEB
pip install fastapi uvicorn # Web API
pip install celery[redis] # Async task queue

VISUALIZATION
pip install matplotlib
```



```
pip install plotly # Interactive charts
pip install pydeck # Deck.gl Python bindings
pip install keplergl # Kepler.gl Jupyter integration

OSM DATA
pip install osmnx # OpenStreetMap data
pip install overpy # Overpass API

FULL INSTALL (one line):
pip install geopandas rasterio shapely pysheds richdem whitebox \
 imdlib xarray netCDF4 osmnx scikit-learn xgboost shap fastapi \
 sqlalchemy geoalchemy2 psycpg2-binary folium plotly pydeck
```